Unit: 6 Unit 2: Exponents Radicals
Topic: Operations with Radicals

Name: ________________
Date: ___________

Homework (Carolina Reaper)

Q1. A musician plays a rhythm with a frequency of $2 \cdot \sqrt{3}$ Hz. If another musician plays a frequency of $\sqrt{3}$ Hz, what is their combined frequency?
(A) $3\sqrt{3}$
(B) $\sqrt{3}$
(C) $2\sqrt{3} + \sqrt{3}$
(D) 3
(E) $2 + \sqrt{3}$

Q2. A painter works on a canvas with a width of $4 \cdot \sqrt{2}$ inches. If the length of the canvas is $2 \cdot \sqrt{2}$ inches, what is the area of the canvas?
(A) 8
(B) 16
(C) $8 \cdot \sqrt{2}$
(D) $4 \cdot \sqrt{2}$
(E) 32

Q3. A sound wave has a frequency of $\sqrt{5}$ Hz. If the frequency is multiplied by $\sqrt{5}$, what is the resulting frequency?
(A) 5
(B) $\sqrt{5}$
(C) $5\sqrt{5}$
(D) $\sqrt{25}$
(E) 25

Q4. A dance troupe performs a routine with a rhythm pattern of $3 \cdot \sqrt{2}$ beats. If the pattern is repeated 2 times, what is the total number of beats?
(A) $6 \cdot \sqrt{2}$
(B) 6
(C) $3 \cdot \sqrt{2}$
(D) 12
(E) $12 \cdot \sqrt{2}$

Q5. A canvas has a length of $\sqrt{7}$ inches and a width of $3 \cdot \sqrt{7}$ inches. What is the area of the canvas?
(A) $3 \cdot 7$
(B) $3 \cdot \sqrt{7}$
(C) 7
(D) 21
(E) $3 \cdot \sqrt{49}$

Q6. A musician divides a rhythm pattern of $8 \cdot \sqrt{3}$ beats by $2 \cdot \sqrt{3}$ beats. What is the resulting rhythm pattern?
(A) 4
(B) 8
(C) 2
(D) 16
(E) $4 \cdot \sqrt{3}$

Q7. A painter mixes $2 \cdot \sqrt{5}$ liters of paint with $3 \cdot \sqrt{5}$ liters of paint. What is the total amount of paint mixed?
(A) $5 \cdot \sqrt{5}$
(B) 5
(C) $2 \cdot \sqrt{5} + 3 \cdot \sqrt{5}$
(D) 10
(E) $10 \cdot \sqrt{5}$

Q8. A sound wave has a frequency of $4 \cdot \sqrt{2}$ Hz. If the frequency is divided by $2 \cdot \sqrt{2}$ Hz, what is the resulting frequency?
(A) 2
(B) 4
(C) $2 \cdot \sqrt{2}$
(D) 8
(E) $4 \cdot \sqrt{2}$

Open Ended Questions (FRQ)

Q9. A dance troupe performs a routine with a rhythm pattern of $x \cdot \sqrt{y}$ beats. If the pattern is repeated z times, what is the total number of beats in terms of x, y, and z?

Q10. A painter works on a canvas with a width of $a \cdot \sqrt{b}$ inches and a length of $c \cdot \sqrt{d}$ inches. What is the area of the canvas in terms of a, b, c, and d?

Chef's Tips

- Tip 1: Emphasize the importance of simplifying radical expressions before performing operations.
- Tip 2: Encourage students to use visual aids such as graphs or diagrams to help with problem-solving.
- Tip 3: Remind students to check their units of measurement when working with real-world applications.